

Computational Physics and Data Science Exercise 3: Freefall

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The goal of this experiment is to fit a linear model to free-fall data of spheres of various radii and material - testing a linear model on a non-linear system containing inherent experimental noise (Height $\pm 10\text{m}$, Time $\pm 1\text{s}$). Data was cleaned of rogue entries (non-numerical, negative, missing values). A linear model was determined on the cleaned data using direct matrix inversion, and a training-test split test performed, yielding $R^2_{\text{training}} = 0.8465$ and $R^2_{\text{test}} = 0.8461$. A correlation matrix was produced to display correlation between features. Then an iterative Stochastic Gradient Descent was performed on adequately scaled data (using `StandardScaler`). This yielded $R^2 = 0.8465$ with the `squared_error` loss function, and $R^2 = 0.8289$ with the `Huber` loss function. The model performed worse for smaller radius spheres (portrayed in a graph of residuals) as these spheres were most influenced by the quadratic drag. The Huber loss function had the biggest variation from the squared error function in the values for radius and mass, showing these values were most affected by the experimental outliers.

I. INTRODUCTION

A sphere falling under gravity is subject to drag. Elementary physics suggests $t \propto \sqrt{h}$, but quadratic drag makes this system much more complicated. The goal of this experiment is to form a model based on many physical features of the system, such as drop height, sphere radius, sphere mass, and air pressure. The data-set consists of experimental data using a video camera of the spheres falling from height. The height measurements have an innate error of $\pm 10\text{m}$ and the timing measurements have $\pm 1\text{s}$. The challenge is fitting an accurate linear model (of form $t = \beta_0 + \sum \beta_i \mathbf{X}_i$) to this process, taking into consideration its non-linear nature, as well as the experimental noise. Two different methods for model fitting will be primarily used. An exact matrix inversion method and an iterative Stochastic Gradient Descent method. Comparisons between the efficacy and robustness of these two methods will be made.

II. THEORY

A body in free-fall with quadratic drag can be modelled using the following equation:

$$t = \sqrt{\frac{m}{kg}} \cosh^{-1} \exp\left(\frac{hk}{m}\right) \quad (1)$$

where $k = \frac{C_d \rho_0 A}{2}$ with the approximation $\rho_0 = \frac{pM}{RT}$. $C_d \approx 0.47$ is the drag coefficient for a spherical object (the objects of interest), A is the cross-sectional area and ρ_0 is the air density. A depends on the radius, m on the spheres density and radius, and ρ_0 on pressure, molar mass, and temperature of the air, as well as the molar gas constant. The variables upon which the time depends are the variables of interest in the CSV file provided for this experiment.

A correlation matrix uses a measure of correlation (in our case, the Pearson correlation coefficient, given:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}} \quad (2)$$

) to compare the correlation between the features (variables) of our data. 1.00 indicates a perfectly positive correlation and -1.00 perfectly negative. These can be easily viewed in a grid. This grid has n^2 possible combinations, but since each variable correlates to itself perfectly, and each correlation of two variables is indicated twice, only $n(n-1)$ independent values are viewed.

The later parts of this equation depend upon Linear Regression methods. The principle and most simple method is the method of least-squares, with loss function

$$\text{MSE} = \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{N} \quad (3)$$

Where y_i are the observed experimental values, $\hat{y}_i$ are the predicted model values, and N is the number of data points. The goal is to minimise the loss function by finding a specific set of values β_i that follow the following equation.

$$t = \beta_0 + \beta_1\rho + \beta_2r + \beta_3m + \beta_4T + \beta_5p + \beta_6h \quad (4)$$

We find these β values by using SciKitLearn to perform the following matrix operation

$$\vec{\beta} = (\mathbf{X}^T \cdot \mathbf{X})^{-1} \cdot \mathbf{X}^T \cdot \vec{y} \quad (5)$$

A second method for linear regression is Stochastic Gradient Descent (SGD). SGD looks to minimise the error by minimising its gradient. This is done by taking steps (of size η , the "learning rate") down of direction of greatest (negative) gradient for random (Stochastic) data points from the entire set. This method is much quicker and less memory intensive but is prone to "zig-zagging" around, and may find itself stuck in local minima rather than the actual minimum value. The gradient function is written explicitly as

$$\nabla_{\beta}\text{MSE}(\beta) = \frac{2}{N}\mathbf{X}^T(\mathbf{X} \cdot \vec{\beta} - \vec{y}) \quad (6)$$

and the β values adjusted as

$$\beta_{\text{new}} = \beta_{\text{old}} - \eta \nabla_{\beta}\text{MSE}(\beta) \quad (7)$$

The SGD method only works for scaled data due to the single learning rate. Scaling all data to roughly the same magnitude prevents the gradient from begin skewed in favour of the large variables. This however creates scaled beta values which must also be unscaled to find the true predictive values.

We can use the Huber Loss function to provide an alternate method for our SGD. The Huber function behaves as MSE for low errors, but above some threshold ε it behaves as $\text{MAE} \propto |\text{Error}|$. This prevents MSE's tendency to overcompensate for large errors and instead allows the model to stay consistent.

III. RESULTS & DISCUSSION

Beginning with the data cleaning and statistics a table (Table I) has been generated to present the maximum, minimum, mean and standard deviations for each feature. 91 lines, representing 1.31% of the data were removed for not following the correct formatting/having bad values.

TABLE I: Descriptive Statistics of Cleaned Data

Feature (Units)	Max	Min	Mean	Std. Deviation
Density (kg/m ³)	7874.0	1200.0	3740.19	2240.92
Radius (m)	0.025	0.005	0.01	0.01
Mass (kg)	0.51535	0.00063	0.09	0.12
Temperature (K)	332.9	257.3	293.12	10.13
Pressure (Pa)	110961.0	87541.0	99935.96	2975.94
Height (m)	1021.0	475.0	749.38	144.91
Time (s)	61.9	9.4	22.22	8.29

Figure 1 is a graph of Fall time (s) vs Drop Height (m) for various radii of magnesium spheres. In this case, magnesium was chosen as a representative plot - all other plots followed very similar trends. We can see that fall time increases both with drop height (as would be expected) but also negatively with radius - that is to say, smaller sphere radii lead to larger fall times. This follows what we would expect: mass depends on radius cubed and drag depends on radius squared. This also leads to the behaviour that the gradient is steeper for smaller radii, as the difference between the mass and drag lessens compared to larger radii. The data has error bars for the constant error provided by our measurement method - ± 1 s for the Fall time and ± 10 m for the drop height.

To quantify how the various features relate to each other, a correlation matrix was generated in order to give a numerical value for how correlated each feature was with each other, allowing us to see particularly which features affected drop time the most.

Focusing on just the features' effect in Figure 2 on the time value, we see that pressure and temperature are effectively zero. This indicates they have little to no effect on fall time. Density and radius both

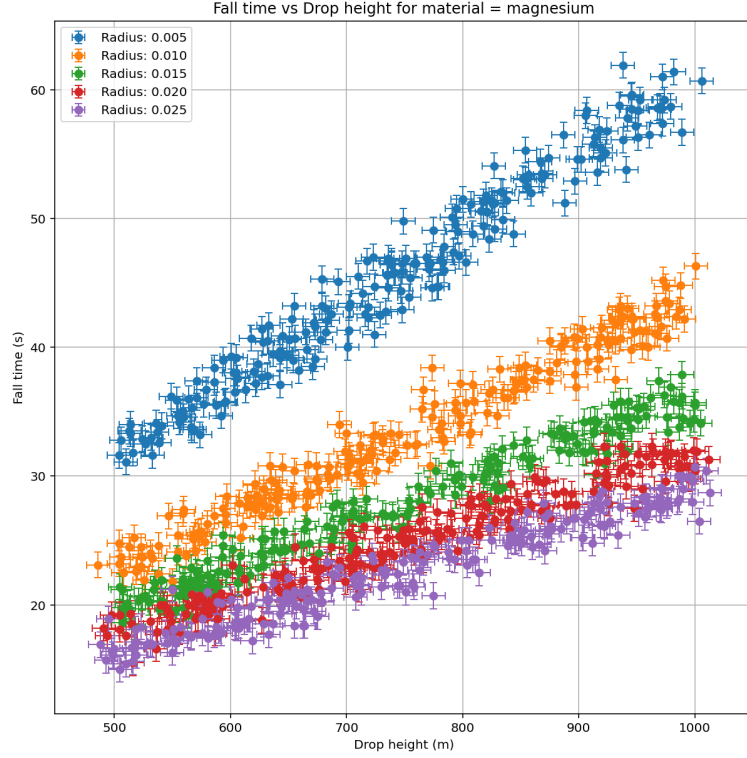


FIG. 1: Fall time vs Height for Magnesium spheres of different radii. This material has been selected as a representative case - all other materials followed the same trends.

have an equal -0.54 correlation with fall time, indicating that increasing density or increasing radius decreases fall time by about the same amount. Mass is also quite well correlated to fall time at -0.50 . This shows an inverse relationship, as theory would suggest. This is potentially troublesome as having all three of these being well correlated suggests a redundancy of sorts - the model may struggle to isolate the effect of mass with density and radius present. The most interesting result is the rather low correlation between drop height and time, namely 0.42 . We would expect from common logic that this be the most correlated pair, but this not being the case suggests that the variations in air resistance (caused by radius and density) and the experimental noise had a considerable effect on the data. This validates the use of multi-variable linear regression, as no single variable dominates.

TABLE II: Fitted β s for the Freefall Data

Coefficient	Feature	Value	Units
β_0	Intercept	27.790	s
β_1	Density	-0.003	s / (kg/m ³)
β_2	Radius	-1059.578	s / m
β_3	Mass	34.647	s / kg
β_4	Temperature	-0.035	s / K
β_5	Pressure	0.000	s / Pa
β_6	Height	0.025	s / m

We can see the values of β generated by the linear regression model on the entire data-set in table II. Some notable values include the fact that β_5 (pressure) is zero. This indicates that variations in pressure have minute effects on the fall time. The value for radius may also seem large, but this is due to the fact that the typical values for radius are small - of order 0.001 . This explains the seemingly large looking value.

Figure 3 shows an issue that comes with this method. Given that the data was trained on the entire dataset, there are some subsets of the data that are represented better by the model. In figure 3a we see that the red dots (the model fit) are below the experimental data. This is due to this particular density and radius combination being "slower" than the model average. This is likely due to greater drag than

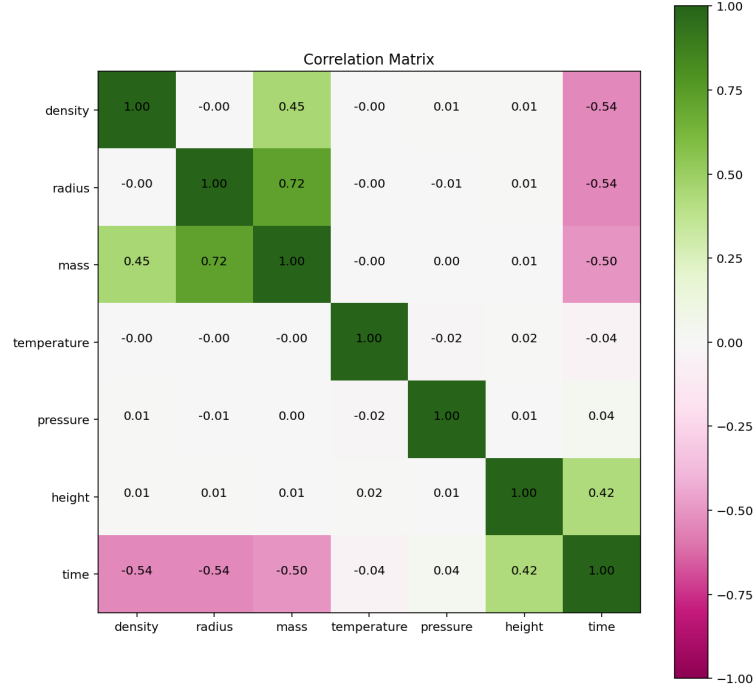


FIG. 2: A correlation matrix between each feature. We see the trivial diagonal of perfect correlation in dark green, as well as areas of negative correlation in pink.

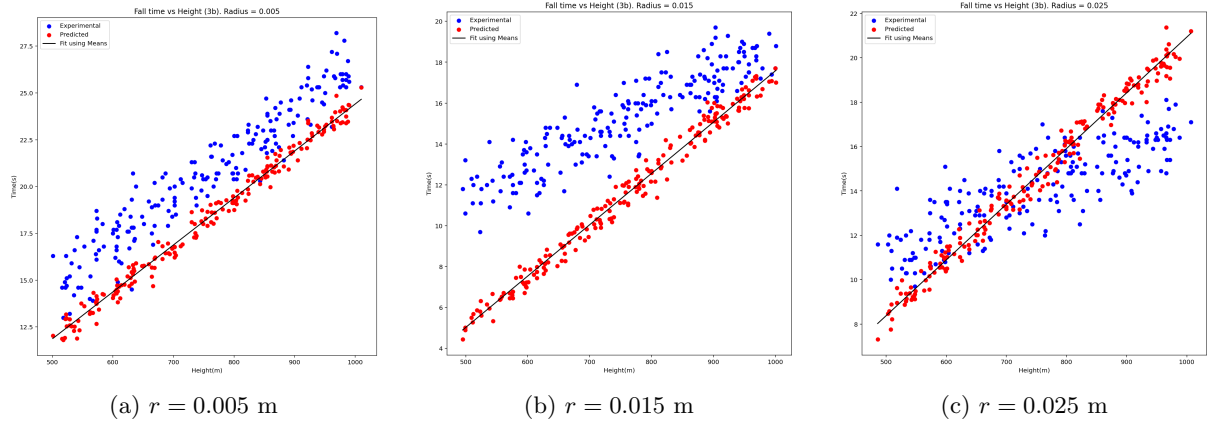


FIG. 3: Comparison of experimental data, modelled predictions and the mean fit for three radii of iron spheres. As the radius increases, the gradient lessens due to influence of drag relative to mass.

average. Despite this, the gradient at this radius is still very similar to the model. Moving through figure 3b to figure 3c we see the gradient of our blue dots decreasing with relation to the model gradient. This is due to the increase in radius causing an increase in mass that overcomes the increase in drag (cubic vs. quadratic increase). The reason that these subsets do not get accurately modelled is because the model itself is too crude to perfectly model free fall. We see in equation 1 that time depends on various non-linear functions (square root, $\cosh^{-1}$, exp). These cause errors in the model. Adding features like $\sqrt{h}$ or Radius $\times$ Density for example can allow the model to more accurately predict the underlying, complex physics.

On splitting the data randomly into 90% training data and 10% test data, we can perform an R^2 test to see if the model is self-assuring. Table III show that the R^2 values are essentially equal - it has captured the relationship very accurately. We can presume that due to the inherent noise provided by the experimental errors that this is a sort of "accuracy limit" of the model. This shows the model is apt for this set of experimental data.

We can see the goodness of fit alternatively through calculating the residuals, i.e. Test Data -

TABLE III: Training-Test Split R^2 values

Dataset	R^2
Training Set	0.8465
Test Set	0.8461
Difference	0.0004

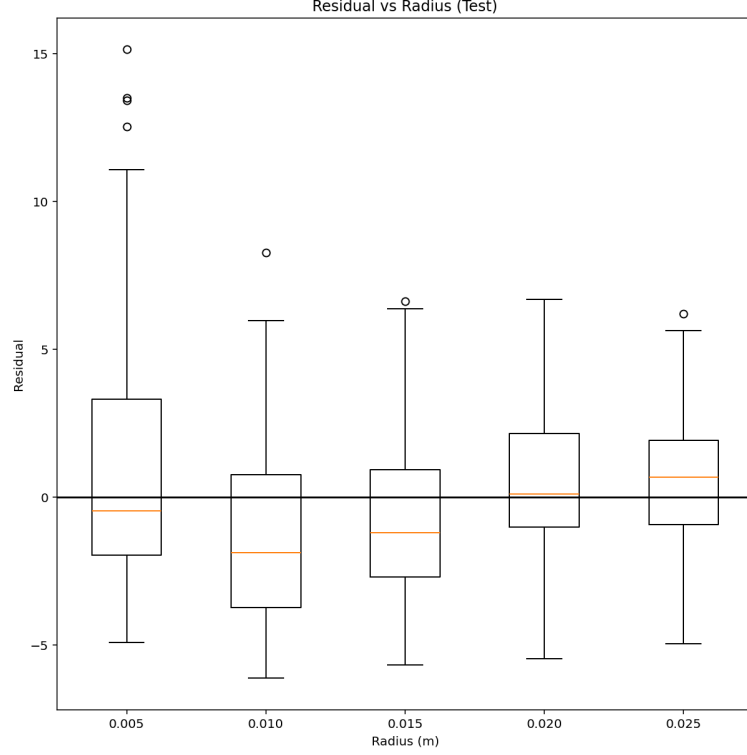


FIG. 4: A box and whisker plot of the residuals (test-prediction) data for each radius. The orange line represents the median, whiskers represent 1.5IQR from Q1/Q3, and the small circles represent outlier data.

Predicted Data. Plotting this as a box plot is a good way to visualise how these residuals lie for individual radii (Figure 4). A clear visible trend is that the length of the box (the interquartile range) decreases with an increase in radius. The whisker length follows a similar trend. We also see more outliers in the smaller radius - showing that a significant number of data points fall outside 1.5IQRs from the third quartile. This is likely due to the inherent noise in the data, stemming from the noted experimental uncertainties on the fall time and the height measurement. For the smaller radii, the median line sits below the zero point. This indicates that the model is over-predicting the fall time. This could be due to the spheres exhibiting non-linear drag (as theory states they will) with the model failing to capture this for smaller spheres. A physical reasoning for the model performing less well for smaller radii is that a smaller radius gives a smaller mass to cross-sectional area ratio (cubic vs quadratic) making the smaller spheres more susceptible to the other parameters such as air density or the quadratic nature of drag.

Running the Stochastic Gradient Descent Regressor on the raw data yielded an R^2 value of $-330669669534007326955862$. Obviously this shows an R^2 far outside the typical range ($0 - 1$). This shows that without appropriate data scaling, the SGD method is completely ineffective, due to the gradients being vast, causing the surface the function is iterating over to begin a long, narrow valley - making it difficult to find the minima. After applying the `StandardScaler` to the data we can more effectively find the actual R^2 value. This is, with the `squared.error` loss function, 0.8465, a value within the accepted range.

The unscaled β values presented in Table IV can be directly compared with the β values obtained using the linear regression model (Table II). We can see easily that the values are very similar between the two models, showing that the SGD and Linear models both converge to the same model. The only value that displays much variance is the intercept. This is likely because SGD is an iterative method

TABLE IV: Unscaled β values from Scaled SGD (`squared_error` loss)

Coefficient	Feature	Value	Units
β_0	Intercept	26.912	s
β_1	Density	-0.003	s / (kg/m ³)
β_2	Radius	-1057.733	s / m
β_3	Mass	34.760	s / kg
β_4	Temperature	-0.032	s / K
β_5	Pressure	0.000	s / Pa
β_6	Height	0.025	s / m

and thus stops as soon as a "good enough" approximation is found, whereas the Linear model is a direct matrix inversion which will find the exact result. This shows that despite SGD not being an exact model, it performs almost equally as well.

If we instead use the **Huber** loss method, we find different results. We can see in Table V that our values differ. Notably in radius and mass. The R^2 value is also slightly lower, at 0.8289. Despite this seeming like the Huber loss function is worse, this deviation may simply be because of the way it works. The Huber model treats large errors linearly. This means that, where `squared_error` will attempt to "tilt" its whole plane to accommodate for large errors, the Huber function will not. It is not skewed by the experimental outliers (which we know exist from Figure 4). The Huber model may in fact be the more physical model - Huber ignores large outliers. The parameters pertaining to sphere diameter appear to be the ones most affected - we can see height is effectively the same. This indicates that in the Linear method, the model may attempt to hyper correct for outliers by increasing the radius value by a large amount to compensate. Radius and mass are also physically linked, meaning the model may struggle to differentiate which one truly causes the affect.

TABLE V: Unscaled β coefficients from Scaled SGD (**Huber** Loss)

Coefficient	Feature	Value	Units
β_0	Intercept	26.416	s
β_1	Density	-0.002	s / (kg/m ³)
β_2	Radius	-870.457	s / m
β_3	Mass	24.791	s / kg
β_4	Temperature	-0.033	s / K
β_5	Pressure	0.000	s / Pa
β_6	Height	0.023	s / m

IV. CONCLUSION

We can see that the Linear model successfully captured the underlying physics, evidenced by the high R^2 values on the data. The residuals were greatest for smaller radii, indicating that the linear model struggles to capture the stronger quadratic drag on these light objects. We can infer that the linear model is only an appropriate model for more massive/greater radius spheres, and begins to break down for small objects, confirming the true non-linear relation that is evident in the theory. The Stochastic Gradient Descent method performed well *after* scaling, showing the importance of scaling all the data to the same order of magnitude to allow the iterative method to be able to calculate reasonable beta values. Once scaling was complete, it was seen that it converged to effectively the same model as the direct matrix inversion, showing that the iterative approach is valid, and a good method to use when matrix inversion becomes too computationally demanding. Then, using the Huber loss function we find a different set of β values. Despite these values differing from the previous analysis; the "true" fit to the data, it can be concluded that the Huber loss function, by scaling large outliers present due to experimental noise linearly rather than quadratically, may have formed a better analysis of the underlying physics.